

ALTIRIG · DESIGN DECISION RECORD

AltiRig design for construction

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/altirig		
Source	docs/decisions/0002-design-for-construction.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-03	Amish Chadha	Constructability review and changes C1 to C17, decided under Amish's 2026-10-03 pre-approvals

0002: Design for construction

- **Date:** 2026-10-03
- **Status:** decided. Amish, 2026-10-03: "start with the first 14 repos from the list of 29 projects. I pre-approve the batch runs along with any recommendations you come up with. I also accept any cost overruns or variations from the assumed scope cost." And, for this batch: "Proceed with the remaining 15 scaffolds". Standard section 18 (Amish, 2026-09-30): "fix the design assumptions to match and be physically feasible".

■ Context

The TRL 2 concept named a vessel, a window, feed-throughs, a stand and baffles but not how any of them is made or held. The constructability review went through every component: how it is made, which faces touch its neighbours, what holds them, and in what order it goes together. The model (`cad/src/model.py`) was checked with `build123d` for overlaps between every pair of parts (none remain) and for clearances (`clearances()`); the masses come from the solids. Nothing below changes what AltIRig does, its pitch or its safety case; the safety-related changes (C3, C15, C16) are all in the conservative direction.

Safety: The vessel is a vacuum vessel. Welding on it is done by a qualified welder, and no part of it is put under vacuum until the safety stops in ALR-BLD-001, section 6, are passed.

■ Changes

Table 1. Changes made for construction.

#	Component	The concept had	The constructable design has	Why
C1	Shell	"A steel drum or tank"	An 813 x 6.35 mm pipe offcut, 1,500 mm long, ends square, holes cut to the nozzle outside sizes	A stocked item; round within 1 % so the buckling calculation holds
C2	Flange rings and door seal	A door with no detail	Two 20 mm plate rings, 960 OD, bored to fit over the pipe and over the door head's skirt; fillet welded both sides; a 6 mm closed-cell EPDM sponge gasket glued to the shell ring's face	Flat cut rings need no machining; the sponge takes up 1 to 2 mm of weld distortion; under vacuum the air closes the joint
C3	Nozzles	A window and feed-throughs in the wall	Heavy-wall set-in stubs (10 in XS for the window, 6 in XS for the feed-through) with square-cut ends standing 16 to 40 mm into the vessel, welded inside and out; no reinforcing pads	No saddle-shaped contour cut on the stub; the heavy wall replaces the metal removed (1.7 and 2.3 times the area needed)
C4	Window	"A polycarbonate or laminated window"	A 340 mm, 20 mm polycarbonate disc on a 6 mm EPDM ring gasket, held by a 10 mm clamp ring over a 25 mm spacer ring with 8 x M10 into the tapped flange	The spacer sets the gasket squeeze (6 to 5 mm) so the window is never over-clamped; no holes in the window
C5	Feed-through	"Sealed feed-through connectors"	A 15 mm aluminium plate with five M20 IP68 glands, cables potted in epoxy for 50 mm, on a 2 mm gasket	Stock glands; potting stops air leaking along the strands
C6	Saddles	Not shown	Two welded saddles: 10 mm web with a curved top, rolled wear plate over 120°, base plate with two anchor holes	Carries the 470 kg vessel at axis height 800 mm with the load spread along the wear plates
C7	Deck	Not shown	Two 50 x 50 x 6 angles welded inside, hanging legs scribed to the wall; an 8 mm aluminium deck bolted to the top legs	The stand needs a flat, removable base; the deck slides in through the door
C8	Baffle	"Flow straightener and baffles"	One 790 mm perforated steel disc, 63 % open, bolted to four welded tabs 590 mm downstream of the propeller	Removable, so the comparison runs can be made with and without it (R6)
C9	Hinge	Not shown	Two 15 mm lugs on each flange ring and a 20 mm pin on the far side; the door lugs' holes slotted 3 mm	Carries the 72 kg door; the slot lets the door seat evenly on the gasket
C10	Latches	Not shown	Three bolt-on latch clamps, at least 3 kN each, at the top, the window side and the bottom	Pull the door into the sponge for the first seal (6.3 kN in all); vacuum does the rest
C11	Thrust stand	"Load cells measuring thrust"	12 mm base plate; two 0.5 mm spring-steel leaves in slit aluminium clamps; anchor block and hanger holding an upright 10 kg bar cell; 10 mm carriage	A bending-beam cell must be loaded across its length: standing it upright lets the carriage push its top end along the axis
C12	Torque head	"Reaction torque"	Housing bored for two 6202 bearings; 15 mm shaft with a hub and an 80 mm motor plate; a	The motor's reaction torque is read without the thrust passing

#	Component	The concept had	The constructable design has	Why
			60 mm arm clamped on the shaft resting on a 5 kg cell on a bracket, through a 3 mm ball	through the torque cell; bearing friction 0.001 N m
C13	Speed sensor	"Optical or electrical RPM"	Optical sensor on an aluminium post on the base plate, 15 mm from the propeller plane, 140 mm below the axis	Fixed to the stand so it moves with nothing; clear of the blades
C14	Valve manifold	"Bleed valve"	A 2 in stub with a tapped 150 mm flange on top; a bought manifold of fittings (relief valve, gauge, bleed and needle valve, vent valve, isolation valve, pump port) bolted on with a gasket	Every valve is a stock fitting; one flange to seal
C15	Bleed valve	Any bleed valve	A normally-open solenoid bleed valve	The chamber vents if power is lost or the emergency stop is pressed (R11)
C16	Feed-through position	Not placed	On the window side, 740 mm from the flange face, 20.9° clear of the propeller plane	Keeps the cable plate out of the fragment zone and close to the control cabinet
C17	Pump and cabinet	Not placed	Pump on the floor on the far side with its exhaust led outdoors; cabinet in front of the door on the window side, so the operator sees the window and reaches the emergency stop	The door swings to the far side; the operator's side stays clear

■ Consequences

- The general arrangement is at Rev P2 (ALR-DWG-001); making sketches ALR-DWG-101 to 122 and the build plan ALR-BLD-001 describe this design.
- The calculations were re-run (ALR-CAL-001 v0.2). The set-in nozzles meet the opening area rule without pads; the stand error budgets are unchanged.
- The bill of materials gains the gaskets, latches, spacer and clamp rings, tabs, hinge and welding labour; the estimated cost is USD 4,920 against the USD 1,000 value-engineering target.
- `design_state: constructable` is set in `project.yaml`.